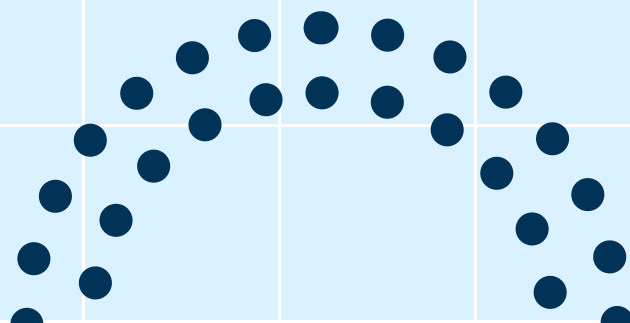
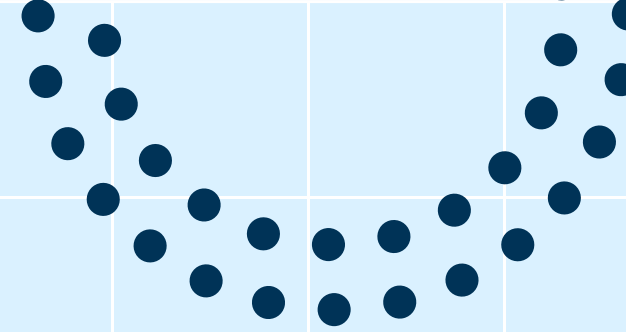
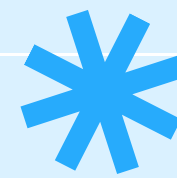


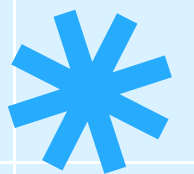
SOLVING TWO-STEP EQUATIONS



What is an Equation?

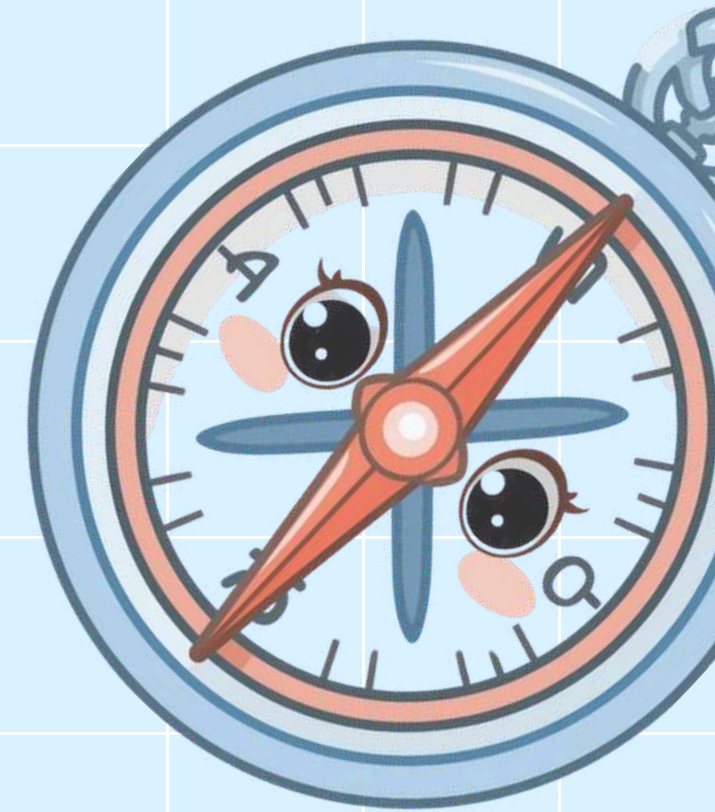


An equation is like a balanced scale - both sides must be equal! Just like $5 + 3 = 8$, whatever is on one side must match the other side.



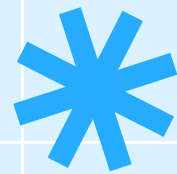
What is a Two-Step Equation?

A two-step equation needs TWO operations to solve! Example: $2x + 3 = 11$. We need to do two things to find the value of x .





Real-Life Example: The Toy Shop



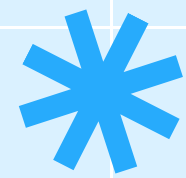
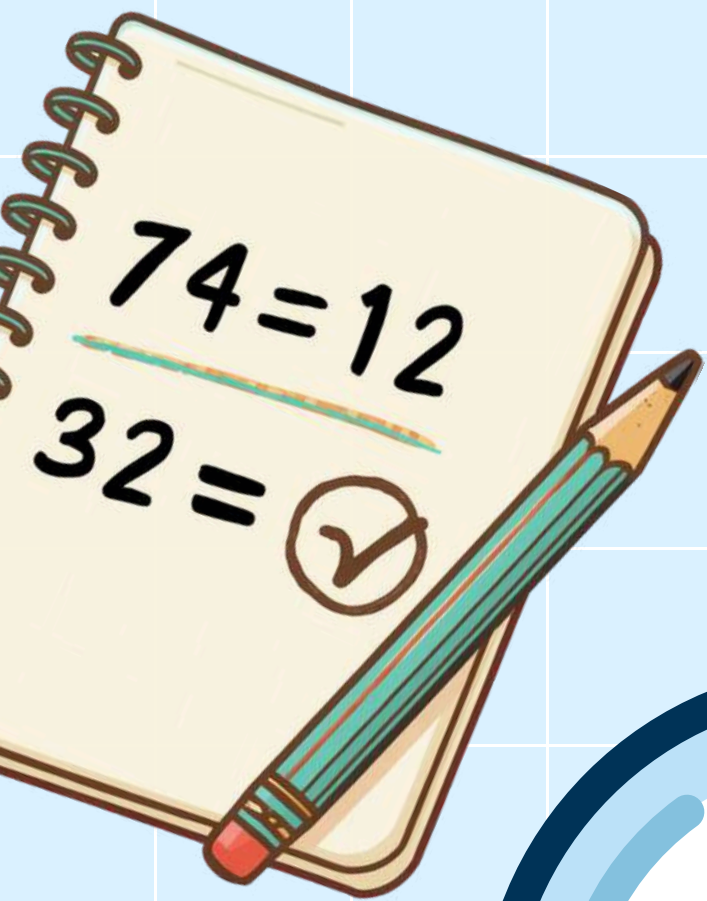
Riya bought 2 toy cars and paid ₹30 extra for gift wrapping. Her total bill was ₹130. Can you find the cost of one toy car? Equation: $2x + 30 = 130$



The Two-Step Strategy



Step 1: Undo addition or subtraction first
Step 2: Undo multiplication or division next
Remember: Always do the OPPOSITE operation to solve!



Example 1 - Let's Solve Together



Solve $3x + 5 = 20$

Step 1: Subtract 5 from both sides $\rightarrow 3x = 15$

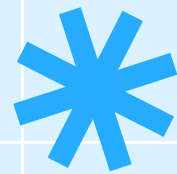
Step 2: Divide both sides by 3 $\rightarrow x = 5$

Check: $3(5) + 5 = 15 + 5 = 20 \checkmark$

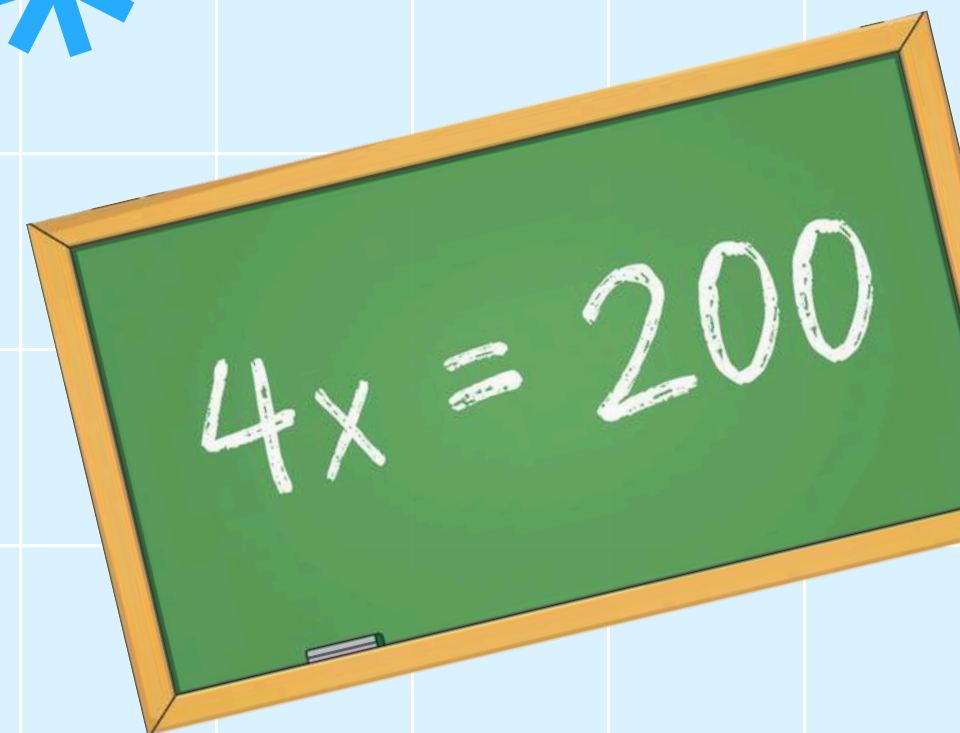




Real-Life Example: The Fruit Seller

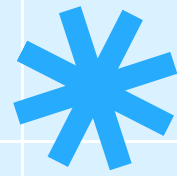


Amit bought 4 mangoes and spent ₹20 on a bag to carry them. His total bill was ₹100. Can you find the cost of one mango? Let's write the equation: $4x + 20 = 100$





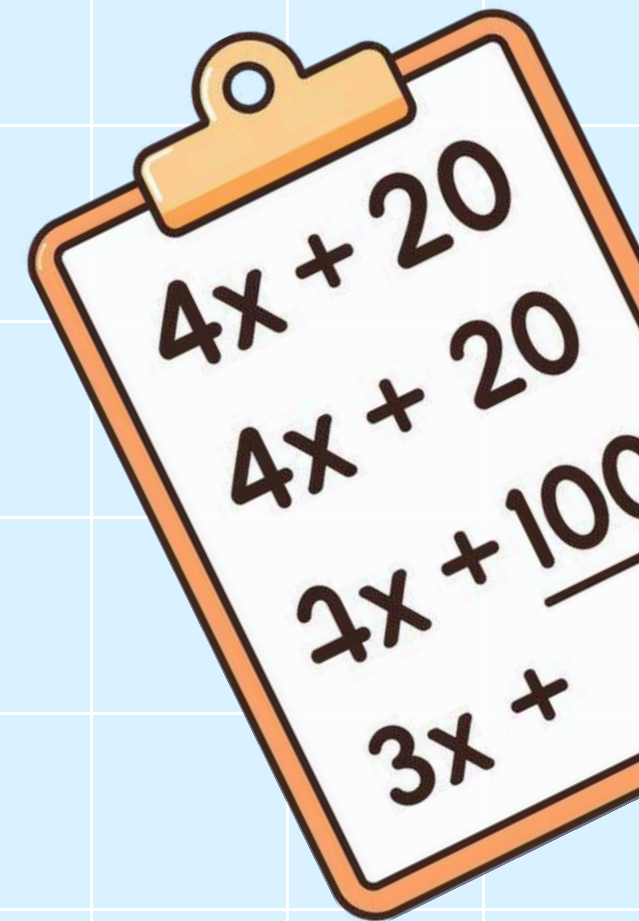
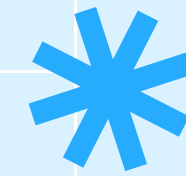
Solving the Fruit Problem



Step 1: Subtract 20 from both sides $\rightarrow 4x + 20 - 20 = 100 - 20 \rightarrow 4x = 80$

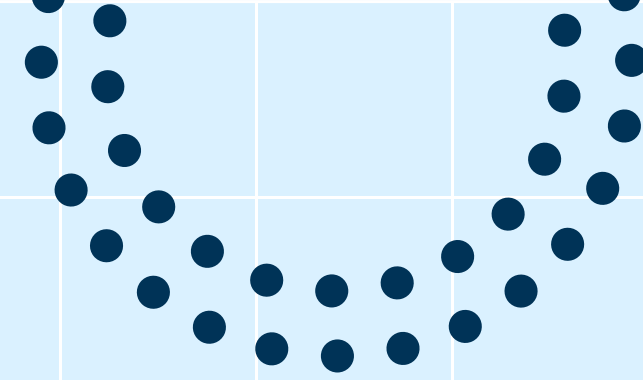
Step 2: Divide both sides by 4 $\rightarrow 4x \div 4 = 80 \div 4 \rightarrow x = 20$

Answer: Each mango costs ₹20! 🍑

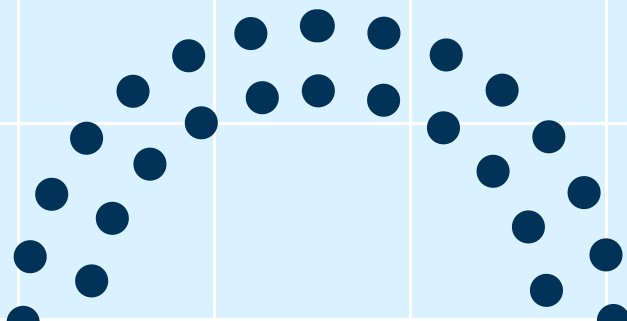




Example 2 - Division First

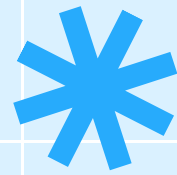


Let's solve $x/2 + 7 = 12$ step-by-step!
Step 1: Subtract 7 from both sides $\rightarrow x/2 = 5$
Step 2: Multiply both sides by 2 $\rightarrow x = 10$
Check: $10/2 + 7 = 5 + 7 = 12 \checkmark$





Real-Life Example - The Birthday Party

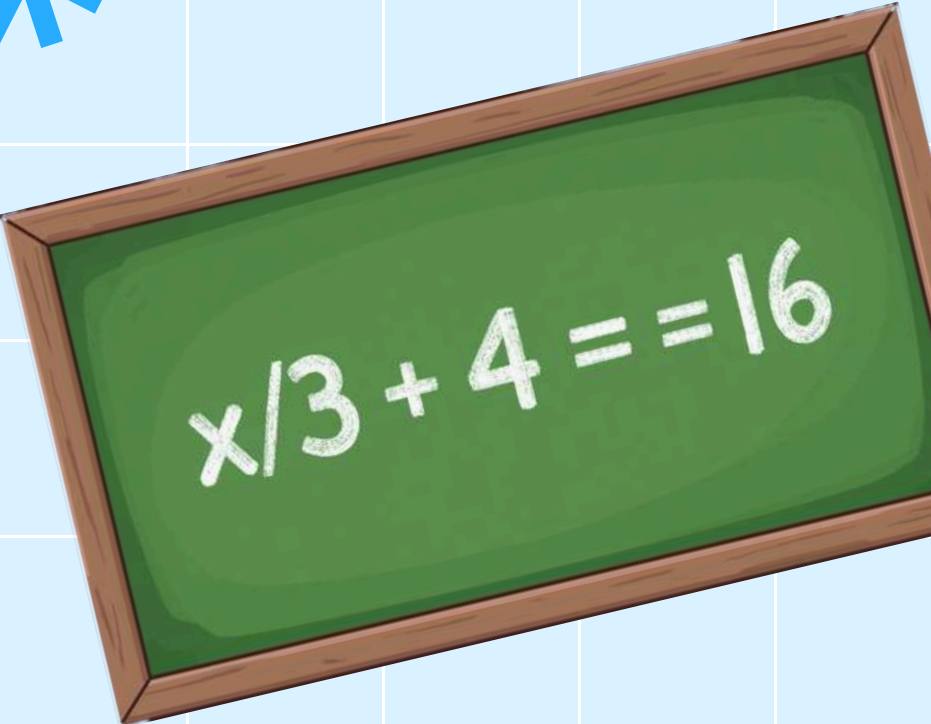


Priya divided her chocolates equally among 3 friends and kept 4 for herself. She had 16 chocolates total.

$$\text{Equation: } x/3 + 4 = 16$$

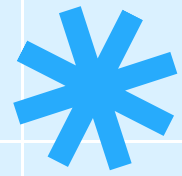
Question: How many chocolates did Priya start with?




$$x/3 + 4 = 16$$



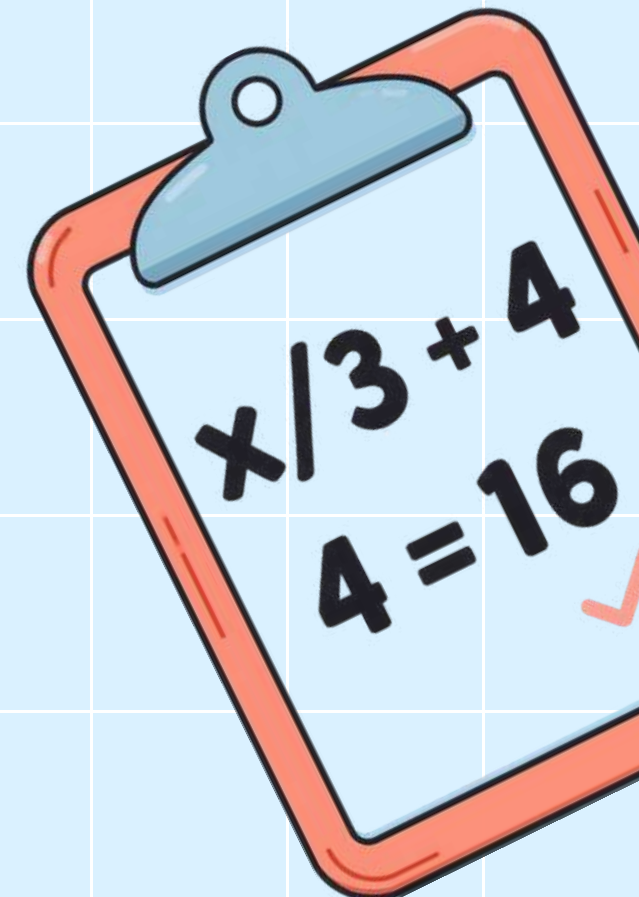
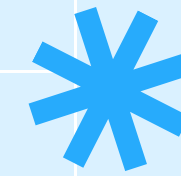
Solving the Birthday Problem



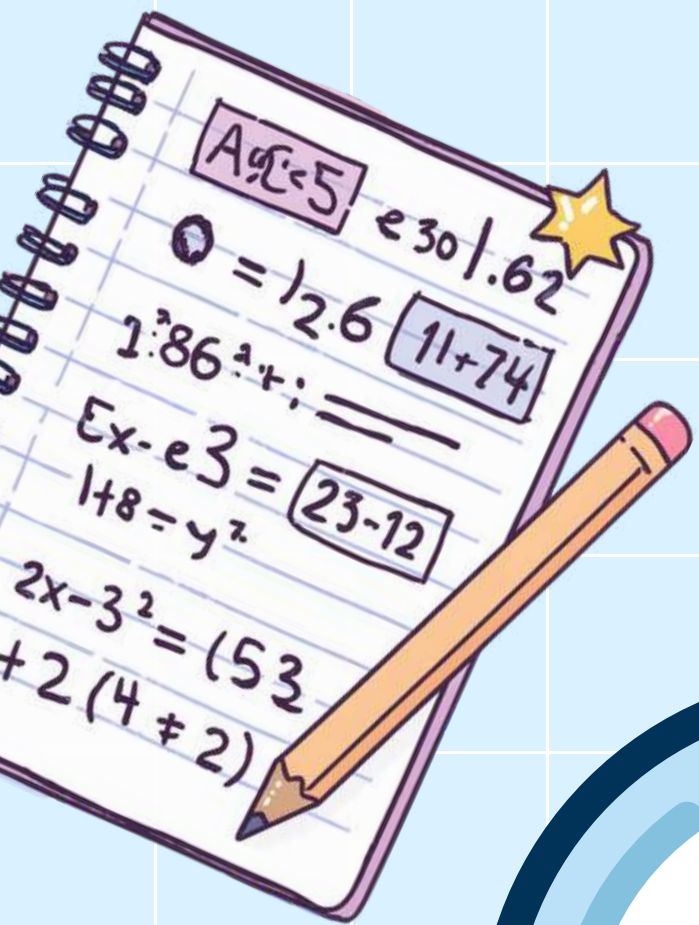
Step 1: Subtract 4 from both sides $\rightarrow x/3 = 12$

Step 2: Multiply both sides by 3 $\rightarrow x = 36$

Priya started with 36 chocolates!



Example 3 - Subtraction Type



$$\text{Solve } 5x - 8 = 17$$

Step 1: Add 8 to both sides $\rightarrow 5x = 25$

Step 2: Divide both sides by 5 $\rightarrow x = 5$

Check: $5(5) - 8 = 25 - 8 = 17 \checkmark$



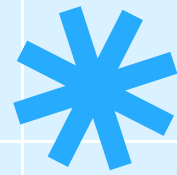
Real-Life Example - The Book Fair

Rohan bought 6 notebooks and got a ₹15 discount. He paid ₹75 in total.
Can you help find the original price of one notebook? Let's write the
equation: $6x - 15 = 75$





Solving the Book Fair Problem



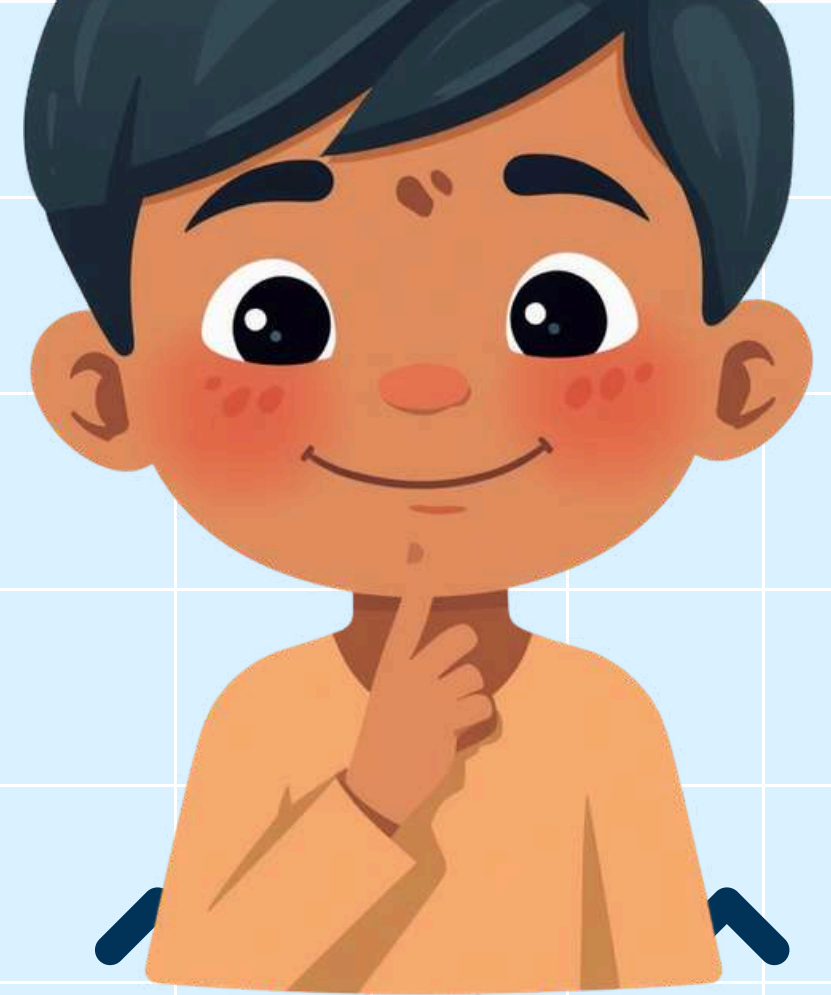
Step 1: Add 15 to both sides $\rightarrow 6x - 15 + 15 = 75 + 15 \rightarrow 6x = 90$

Step 2: Divide both sides by 6 $\rightarrow 6x \div 6 = 90 \div 6 \rightarrow x = 15$

Answer: Each notebook originally cost ₹15!



Practice Time!



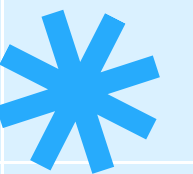
Try these on your own! Test your two-step equation skills:

- 1.
- 2.
- 3.

$$2x + 6 = 14$$

$$4x - 3 = 13$$

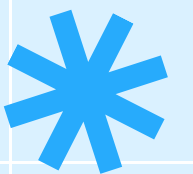
$$x/5 + 2 = 7$$



Answers & Tips

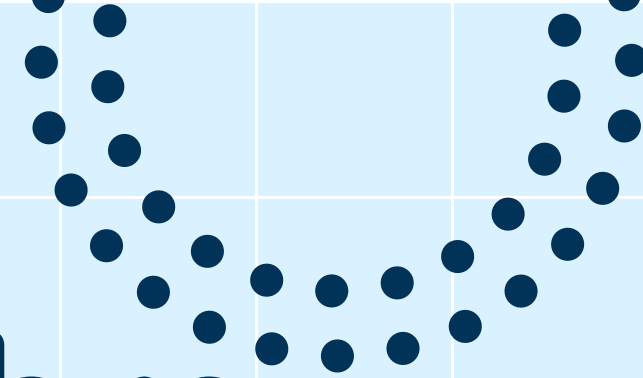


1. $x = 4$ 2) $x = 4$ 3) $x = 25$
2. Remember: Always check your answer! Do the opposite operation and keep the equation balanced.





You're an Algebra Star!



Great job learning two-step equations! Practice makes perfect. Keep solving and have fun with math! You've mastered the art of balancing equations like a true mathematician!

